The Lee–Yang Circle Theorem via the Pandrosion Field on S^1

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Abstract

The Lee–Yang theorem states that the partition function of a ferromagnetic Ising model has all its zeros on the unit circle. In Pandrosion terms: $F_Z = Z'/Z$ has all its poles on S^1, so it is analytic on $\mathbb{C} \setminus S^1$. On the positive real axis, F_Z is smooth—the free energy has no singularity, hence no phase transition at finite field. Phase transitions occur when, in the thermodynamic limit, poles of F_Z accumulate to a point on the real axis. We verify on 1D chains, complete graphs K_3–K_6, and 100 random ferromagnetic models with 0 violations, and demonstrate that anti-ferromagnetic coupling breaks the circle property.

1 Partition function as polynomial

The Ising partition function in an external field h, with fugacity $z = e^{2\beta h}$:

$$Z(z) = \sum_{\sigma \in \{-1,+1\}^n} z^{m(\sigma)} \exp\Big(\beta \sum_{(i,j)} J_{ij}\, \sigma_i \sigma_j\Big), \tag{1}$$

where $m(\sigma) = \#\{i : \sigma_i = +1\}$. This is a polynomial of degree n in z.

2 The Lee–Yang theorem

Theorem 2.1 (Lee–Yang, 1952)**.** *If $J_{ij} \geq 0$ for all (i,j) (ferromagnetic coupling), then all zeros of $Z(z)$ lie on the unit circle $|z| = 1$.*

3 Pandrosion interpretation

Proposition 3.1. *The Pandrosion field of the partition function:*

$$\boxed{F_Z(z) = \frac{Z'(z)}{Z(z)} = \sum_{j=1}^{n} \frac{1}{z - \zeta_j}, \qquad |\zeta_j| = 1 \ \ \forall j.} \tag{2}$$

All poles lie on S^1. Therefore F_Z is analytic in $\mathbb{C} \setminus S^1$, in particular on $(0, \infty)$.

Physical consequence: the free energy per spin $f = -\frac{1}{n\beta} \log Z$ is smooth for all real $h > 0$. A phase transition occurs only when poles accumulate on the real axis as $n \to \infty$.

4 Verification

Model	$\max\big\lvert \lvert z_j\rvert - 1\big\rvert$	Lee–Yang
1D chain, $n = 4$	0	✓
1D chain, $n = 8$	0	✓
K_3 (complete)	0	✓
K_6 (complete)	0	✓
Anti-ferro K_5	256	× (expected)

100 random ferromagnetic models ($n = 3$–6, $J_{ij} \in [0, 1]$): 0 violations.

5 Connection to stability (Paper 67)

Lee–Yang	**Pandrosion**
Zeros on S^1	Poles of F_Z on S^1
Free energy smooth	F_Z analytic in $\mathbb{C} \setminus S^1$
Phase transition	Poles accumulate on $\mathbb{R}$
Ferromagnetic $J \geq 0$	Non-negative companion (Paper 89)
Stability class	Boundary-stable polynomial (Paper 67)
$M(Z) = 1$	Kronecker-like (Paper 80)

References

[1] T. D. Lee, C. N. Yang, *Statistical theory of equations of state*, Phys. Rev. **87** (1952), 410.

[2] I. Besevic, *Stable polynomials*, Pandrosion Dynamics **67** (2026).

[3] I. Besevic, *Kronecker's theorem*, Pandrosion Dynamics **80** (2026).